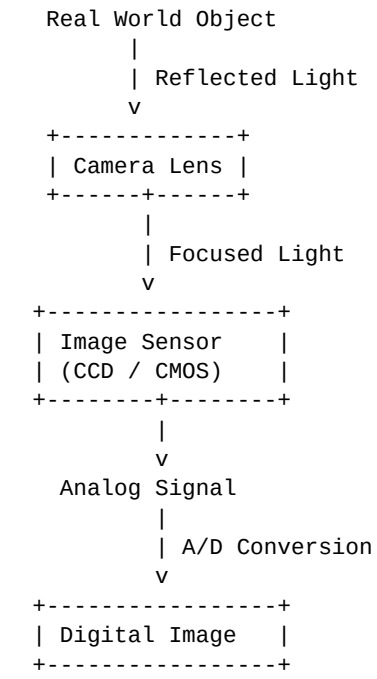
Computer Vision – Assignment 1

Q1. Fundamentals of Image Formation with Example

Answer

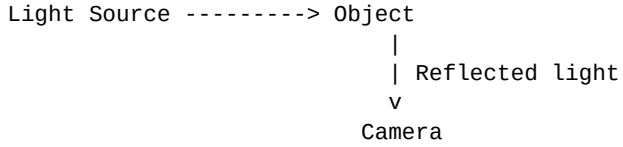
Image formation is the process by which a real-world 3D scene is captured by a camera and represented as a 2D digital image. In computer vision, it describes how light from an object travels through the camera lens and reaches the image sensor.

Basic Image Formation Model



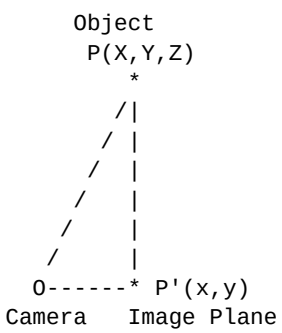
1. Illumination

First, an object is illuminated by a light source such as the Sun, bulb, or LED. The object reflects some of this light toward the camera. The amount of reflected light depends on the object's surface properties and the illumination.



2. Camera Projection

The camera receives reflected light through its lens. A simple camera can be modeled using the pinhole camera model.



Using similar triangles:

$$x = fX / Z$$

$$y = fY / Z$$

where (X, Y, Z) are the 3D coordinates of an object point, (x, y) are the corresponding 2D image coordinates, f is the focal length, and Z is the distance of the object from the camera.

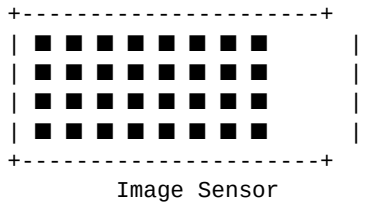
Important observation: As Z increases, the projected object becomes smaller.

$$\text{Image size} \propto 1 / Z$$

Therefore, distant objects appear smaller in an image.

3. Image Sensor

The camera sensor, usually CCD or CMOS, converts incoming light into an electrical signal. The sensor contains a large number of small elements called pixels. Each pixel records the intensity of light falling on it.



4. Sampling and Quantization

Sampling: Sampling converts the continuous spatial image into discrete pixels. If an image has M × N pixels, it contains M × N discrete samples.

Quantization: Quantization converts continuous intensity values into discrete numerical values. For an 8-bit grayscale image:

$$L = 2^{8} = 256$$

Therefore, intensity values range from 0 to 255, where 0 represents black, 255 represents white, and intermediate values represent shades of gray.

5. Digital Image Representation

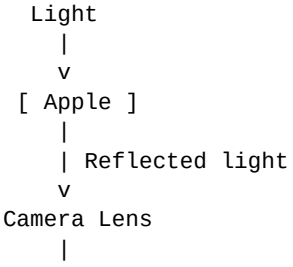
A grayscale image can be represented mathematically as f(x, y), where x is the horizontal pixel coordinate, y is the vertical pixel coordinate, and f(x, y) is the intensity value at that pixel.

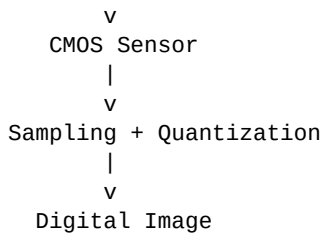
$$f(x,y) = \begin{bmatrix} 0 & 50 & 100 \\ 80 & 150 & 200 \\ 120 & 220 & 255 \end{bmatrix}$$

Every number in the matrix represents the intensity of one pixel.

Example: Image of a Red Apple

Suppose a camera captures an image of a red apple. The image formation process can be summarized as:





The camera records the amount of light reflected from different parts of the apple. The resulting digital image contains information about the apple's shape, color/intensity, edges, texture, and background. These pixel values can then be processed for object detection, recognition, segmentation, and tracking.

Factors Affecting Image Formation

- **Illumination** – amount and direction of light.
- **Object reflectance** – how much light the object's surface reflects.
- **Camera parameters** – focal length, aperture, shutter speed, etc.
- **Distance from camera** – affects apparent object size.
- **Sensor characteristics** – resolution and sensitivity.
- **Sampling and quantization** – determine digital image quality.

Conclusion

Image formation is the process of converting a 3D real-world scene into a 2D digital image. It involves illumination, reflection, camera projection, sensing, sampling, and quantization. Understanding image formation is fundamental to computer vision because almost every subsequent operation works on the resulting digital image.